Measuring Teacher Attrition with Labour Force Survey Data

The Case of the United States

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Why this project

- Research on the decision to leave teaching needs a person-level leave-or-stay outcome linked to rich covariates: family events, pay, hours, pension coverage, destinations after exit.
- Administrative records count leavers best, but they are often inaccessible, cover one state and the public sector only, and carry few covariates.
- Large household surveys are the accessible alternative. Using them well requires knowing the measurement choices they impose — and what each choice does to the estimate.

Contribution

1. **A user's guide.** A systematic account of the choices involved in constructing a leaving rate from a longitudinal household survey — linking individuals, defining a leaver — the error each choice introduces, its direction, and how much it moves the estimate.

2. **An implementation.** The guide applied to twenty-eight years of the Current Population Survey, benchmarked against the national reference on both the composition of the workforce and the rate.

3. **A demonstration.** What a survey-based outcome permits that records rarely do: the correlates of leaving, the destinations of leavers, and the heterogeneity across states.

Where do the official statistics come from?

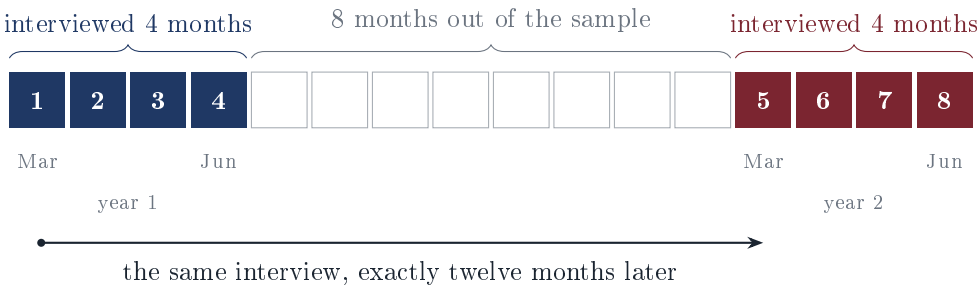
- The national reference is the NCES Teacher Follow-up Survey: itself a sample survey, run in occasional waves — eight since 1988–89 — and published with a lag.
- Truly administrative data exist only state by state (longitudinal data systems, pension registers): public sector only, and they measure exit from the state school system, not from the profession.
- There is no national administrative count. Every national number is survey-based; the question is how the survey is used.

The CPS replicates the official picture — and does it every year

	CPS	NTPS (official)	Difference
Measurement	**every year, 1997–2024**	occasional waves	
Example wave: 2020–21, the latest NTPS			
Female (%)	75.5	76.8	−1.3
Master's degree or higher (%)	57.0	61.0	−4.0***
Average age (years)	44.0	42.9	+1.1***
Teachers observed in the wave	3,855	39,633	

Public school teachers. CPS: supplements of 2021–2022 (school years 2020–21 and 2021–22), weighted. NTPS: National Teacher and Principal Survey 2020–21 (NCES 2022-113; Digest of Education Statistics, table 209.10; respondent count from NCES 2024-024). Stars test the CPS mean against the published NTPS estimate with design-adjusted standard errors: *** $p < 0.001$; the female share does not differ significantly.

Full sample: **5.0 million** person-year records, 1998–2025; **104,545** teacher observations; **48,842** teachers followed across interviews in the linked panel.


interviewed 4 months
8 months out of the sample
interviewed 4 months
1
2
3
4
5
6
7
8
Mar
Jun
Mar
Jun
year 1
year 2
the same interview, exactly twelve months later

The March supplement (ASEC)

- The largest supplement of the CPS: the March sample is expanded to roughly 90,000 households.
- Every adult reports on the *previous calendar year*: longest job held, weeks worked, earnings, benefits.
- We use it to reconstruct the leaving rate.

Who counts as a teacher

Occupation of the longest job held last year, CPS ASEC — three Census classifications, 28 survey years

Surveys 1998–2002 teaching years 1997–2001 *1990 Census classification*		Surveys 2003–2019 teaching years 2002–2018 *2002/2010 Census classification (SOC)*		Surveys 2020–2025 teaching years 2019–2024 *2018 Census classification (SOC 2018)*	
155	Teachers, prekindergarten and kindergarten	2300	Preschool and kindergarten teachers	2300	Preschool and kindergarten teachers
156	Teachers, elementary school†	2310	Elementary and middle school teachers†	2310	Elementary and middle school teachers†
157	Teachers, secondary school†	2320	Secondary school teachers	2320	Secondary school teachers
158	Teachers, special education	2330	Special education teachers	2330	Special education teachers
159	Teachers, n.e.c. (not elsewhere classified)	2340	Other teachers and instructors	2360	Other teachers and instructors

Never included (they carry their own codes, outside the set): postsecondary teachers (2200s) · tutors (2350, 2018 cl.) teaching assistants (2540/2545) · childcare workers (4600) · education administrators (0230)

Are the codes traceable across census years?

Yes as a block: the K–12 teaching group maps one-to-one across the three classifications. Every figure uses the union of the codes, never a sub-code across regimes.

† sub-codes are not: middle-school teachers sit under elementary or secondary before 2003 and with elementary (2310) after; "other teachers" is renumbered 2340 → 2360 in 2020.

Yes within each survey: last year's job (OCCUP) and the current job (PEIOOCC) share one classification, so no leaver/switch comparison straddles a code change (seam surveys: 2003 and 2020).

Yes for people, but only two years: the CPS links the same person across two consecutive March interviews (PERIDNUM), never longer.

Two ways to measure the leaving rate

Follow the person, year to year

The rotation shows the same individual twelve months apart. The CPS has no person identifier, so records are linked on fields common to both years:

household identifier
dwelling identifier
person line number within the household

validated on sex, race,
and age advancing 0–2 years

Ask about last year, each March

The supplement's retrospective question. Whoever reports teaching as last year's longest job, and no longer teaches at the March interview, left within the year.

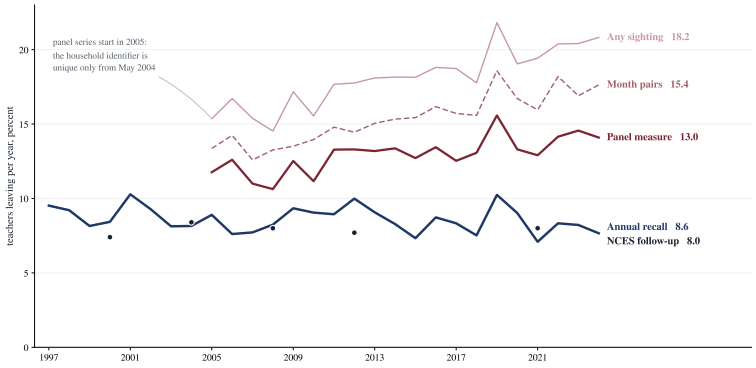
The measure used in the literature so far.

Measuring the leaving rate: the alternatives, side by side

Measure	A teacher is...	A leaver is...	Rests on...
Annual recall	whoever taught as last year's longest job	no longer teaching at the March interview	accurate recall of the previous year
Occupation pair	a teaching occupation code one year ago	a different occupation code today	a current code existing; fails for the non-employed
Panel (benchmark)	teaching in two or more year-1 interviews	never teaching in year 2, given two or more interviews, one outside the summer	correct linkage across waves; movers cannot be linked
Month pairs, any sighting	every teaching month, or a single sighting	not teaching twelve months after each month	one person counting several times; admits one-month spells

Same records throughout; only the definition changes. Each row answers the same three questions.

One dataset, one rate per definition



Message. The definition of a leaver moves the measured rate by a factor of nearly four.

Policy. Comparisons across studies and countries must hold the definition fixed.

About 3,700 teachers observed per year in the recall series; about 2,500 classified per year in the panel measures. Dots: NCES Teacher Follow-up Survey.

Why the lines differ

18.2 Any sighting. One monthly sighting admits substitutes and one-month spells: of teachers seen teaching in a single year-1 interview, 55 percent are never seen teaching in their year-2 interviews, against 11 percent of those seen in all four. A single month of occupation coding is noisy; summer-only patterns are just 3 percent of these leavers.

15.4 Month pairs. The unit is the pair, not the person: the same teacher can count as a leaver in one pair and a stayer in another.

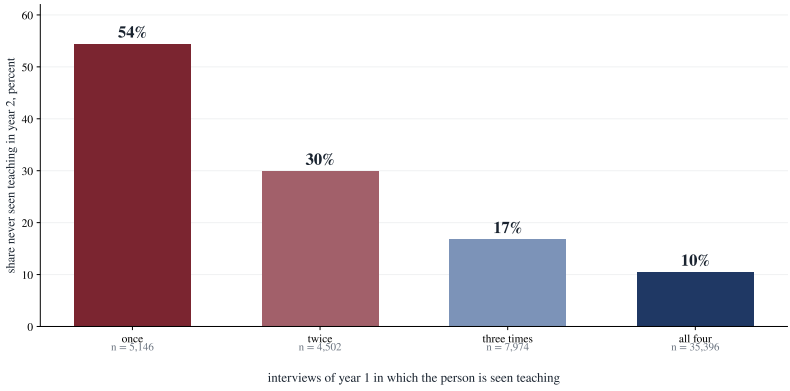
13.0 Panel measure. One verdict per person. Still a point-in-time notion of the year, and one leaver in six is teaching again within three months.

8.6 Annual recall. The main job of the year, revisited a year later — the concept of the official survey, which is why it lands next to the dots. Its own risk is recall error.

8.0 NCES follow-up. The reference wave.

Each construction answers a slightly different question; the choice, not the data, drives the spread.

The attachment gradient, shown



Message. Who the denominator admits drives the top lines: seen teaching once, 54 percent are never seen teaching a year later; seen all four times, 10 percent.

Caveat. Linkage follows addresses, not people: movers cannot be followed, and movers leave more, so the panel measures are lower bounds on mobility-related exit.

Why teachers leave: what the CPS can and cannot see

Reasons identified in the literature	Observed in the CPS?
Working conditions	no
Organizational support (mentoring, leadership)	no
Autonomy and intrinsic motives	no
Financial factors	**yes**
Personal life-cycle motives	**yes**
Sociodemographic characteristics	**yes**

Five broad families of reasons from our review of the literature. The CPS covers the economic and life-cycle margins.

Five messages

1. Economic and employment conditions correlate with leaving far more than demographics.

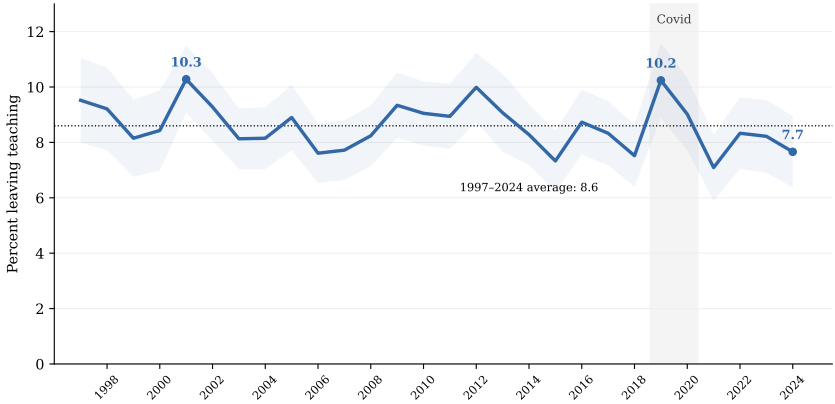
2. Exits concentrate where attachment and annual pay are lowest: part-time and part-year teachers in the bottom of the earnings distribution.

3. Those same exits lead *out of employment*, not to better-paid jobs: two thirds of bottom-quintile leavers exit the labor force. Low pay and non-employment destinations are one phenomenon, not two.

4. Life-cycle events are the largest single trigger: childbirth raises a woman's exit probability by about six points, more than in comparable professions.

5. Pension coverage retains, and attrition differs widely across states: the actionable margins are contractual and institutional.

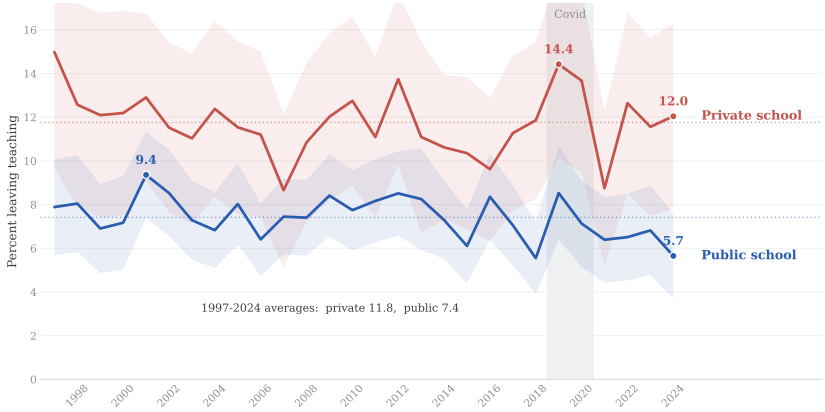
Leaving teaching, 1997–2024



Message. The rate fluctuates around 8.6 percent with no trend: a persistent outflow, not a mounting exodus.

Policy. Retention faces a steady-state cost, not an emergency spike.

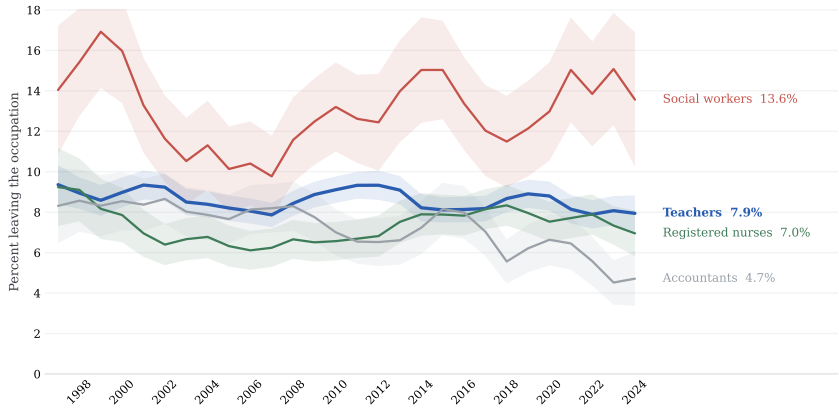
Leaving teaching, by school sector



Message. Private schools lose teachers at roughly twice the public rate.

Policy. Retention tracks contract structure, pay and pensions, not classroom conditions alone.

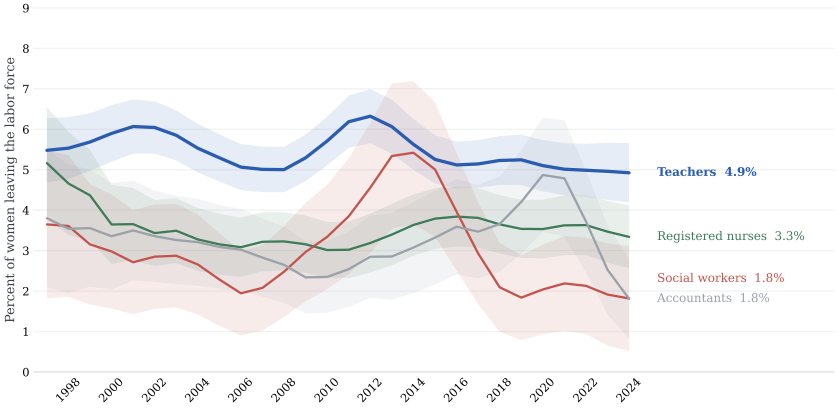
Leaving the occupation, by profession



Message. Teacher attrition is unexceptional next to comparable professions: above nurses and accountants, well below social workers.

Policy. Benchmarks from other professions discipline how alarming any teacher figure should be.

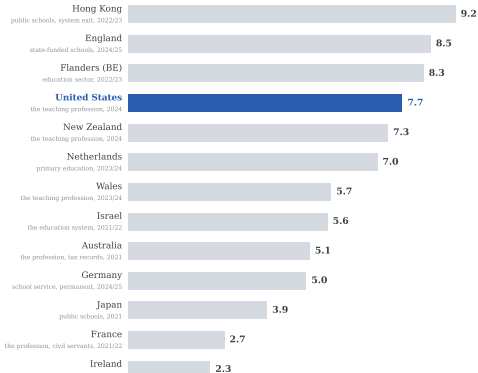
Labor force exit of women, by profession



Message. Women teachers exit the labor force at about five percent a year, well above nurses, social workers and accountants.

Policy. The family-compatibility margin is specific to teaching, not to professional work in general.

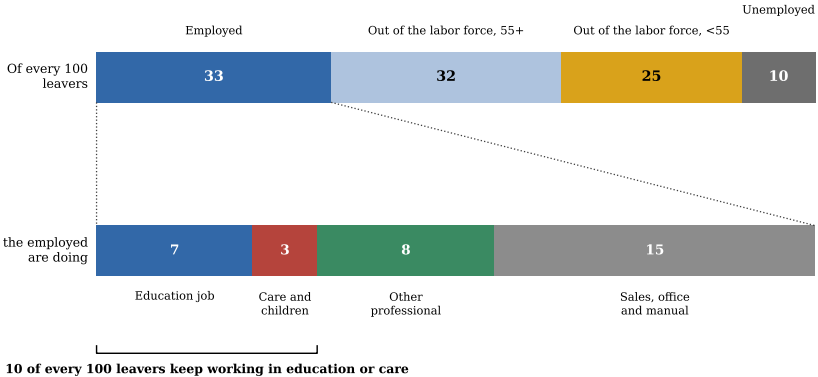
Teachers leaving per year, across countries



Message. The US sits in the upper range of the school systems that publish a rate.

Policy. Comparable, survey-based measurement is what makes this ranking possible.

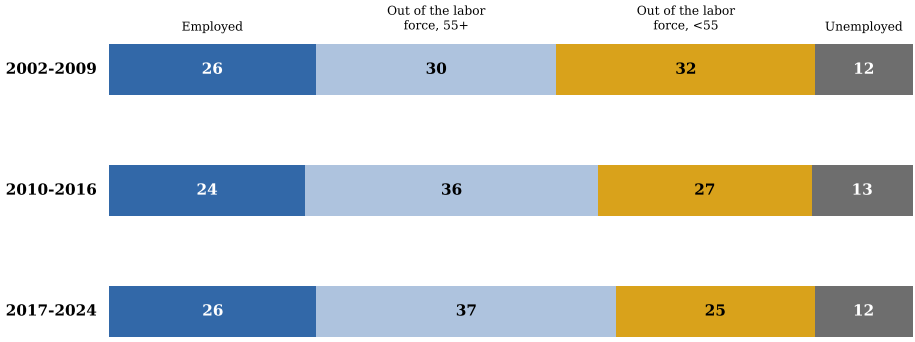
Destinations of teachers who leave, 2024



Message. Of every hundred leavers, about a third take another job; most exit employment altogether.

Policy. Competing employers are not the main counterfactual; retirement and care are.

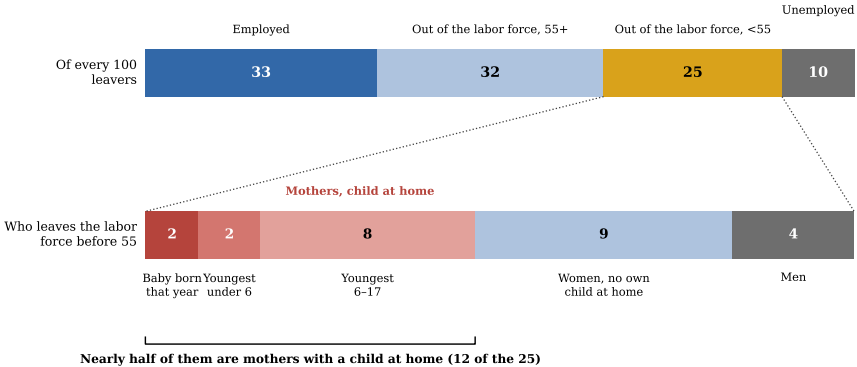
The routes out of teaching, 2002–2024



Message. The destination mix has been essentially stable for two decades.

Policy. The composition of the outflow is structural, not cyclical noise.

Who leaves the labor force before 55



Message. Nearly half of the pre-55 labor-force exits are mothers with a child at home.

Policy. Childcare and parental-leave policy are retention policy.

Where the employed leavers go: every occupation

EDUCATION, NON-TEACHING — 23%

8.1	Postsecondary teachers
4.3	Tutors
4.2	Teaching assistants
3.1	Education administrators
1.5	Librarians
1.3	Other instruction and library wo...
0.2	Archivists, curators, and museum...

CARE — 9%

1.8	Educational and career counselors
1.2	Social and human service assista...
1.1	Mental health and substance abus...
1.1	Social workers, all other
0.8	Childcare workers
0.6	Mental health counselors
0.5	Counselors, all other
0.5	Substance abuse and behavioral d...
0.4	Other community and social servi...
0.4	Probation officers and correctio...

PROFESSIONAL & MANAGERIAL — 32%

9.7	Managers, all other
1.9	Coaches and scouts
1.7	Architects
1.5	General and operations managers
1.3	Business operations specialists...
1.2	Other mathematical science occup...
1.1	Writers and authors
0.9	Broadcast, sound, and lighting t...
0.8	Chemical technicians
0.7	Social and community service man...
0.6	Property, real estate, and commu...
0.6	Photographers
0.6	Actors
0.6	Therapists, all other
0.5	Lodging managers
0.5	Construction managers
0.5	Miscellaneous health technologis...
0.4	Other psychologists
0.4	Other engineering technologists...
0.4	Management analysts
0.4	Market research analysts and mar...
0.4	Human resources workers
0.4	Veterinarians
0.4	Chief executives
0.4	Dietitians and nutritionists
0.4	Information security analysts
0.3	Claims adjusters, appraisers, ex...
0.3	Physical therapists
0.3	Food service managers
0.3	School psychologists
0.3	Editors
0.3	Transportation, storage, and dis...
0.2	Meeting, convention, and event p...
0.2	Training and development managers
0.2	Accountants and auditors
0.2	Other healthcare practitioners a...
0.2	Graphic designers
0.2	Training and development special...
0.2	Materials engineers
0.2	Civil engineers
0.1	Other designers
0.1	Wholesale and retail buyers, exc...
0.1	Physical scientists, all other
0.1	Project management specialists
0.1	Dietetic technicians and ophthal...
0.0	Floral designers
0.0	Farmers, ranchers, and other agr...
0.0	Landscape architects
0.0	Miscellaneous social scientists...

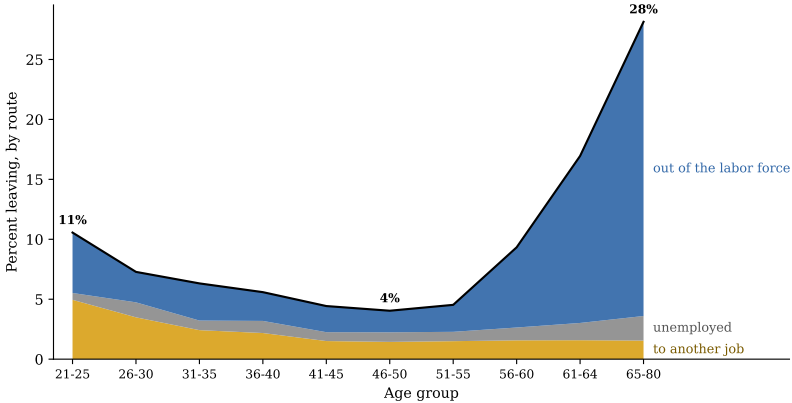
SALES, OFFICE & MANUAL — 36%

2.8	Secretaries and admin. assistants
1.8	Retail salespersons
1.8	Landscaping and groundskeeping w...
1.6	Exercise trainers
1.4	Supervisors of personal care and...
1.3	Taxi drivers
1.3	Police officers
1.2	Executive secretaries
1.1	Stockers and order fillers
1.0	Other life, physical, and social...
1.0	First-Line supervisors of non-re...
0.9	Title examiners, abstractors, an...
0.8	Personal care and service worker...
0.7	Fast food and counter workers
0.7	First-Line supervisors of retail...
0.7	Couriers and messengers
0.7	Statistical assistants
0.7	Paralegals and legal assistants
0.6	Cashiers
0.6	Recreation workers
0.6	First-line supervisors of landsc...
0.6	Legal support workers, all other
0.6	Other healthcare support workers
0.6	Laborers and freight, stock, and...
0.6	Animal caretakers
0.5	Insurance claims and policy proc...
0.5	Personal care aides
0.5	Bill and account collectors
0.4	Waiters and waitresses
0.4	Data entry keyers
0.4	Hairdressers, hairstylists, and...
0.4	Real estate brokers and sales ag...
0.4	Customer service representatives
0.4	First-Line supervisors of office...
0.4	Manicurists and pedicurists
0.4	Other entertainment attendants a...
0.4	Construction laborers
0.4	Office and administrative suppor...
0.3	Cooks
0.3	Bookkeeping, accounting, and aud...
0.3	Miscellaneous agricultural worke...
0.3	Travel agents
0.3	Sewing machine operators
0.3	Bartenders
0.3	Conveyor, dredge, and hoist and...
0.3	Boilermakers
0.3	Billing and posting clerks
0.3	Occupational health and safety s...
0.3	Packaging and filling machine op...
0.2	Laundry and dry-cleaning workers
0.2	Physical therapist assistants an...
0.2	Janitors and building cleaners
0.2	Payroll and timekeeping clerks
0.2	Nursing assistants
0.2	Helpers–production workers
0.1	Hosts and hostesses, restaurant...
0.1	Sales representatives, wholesale...
0.1	Security guards and gambling sur...
0.1	Chefs and head cooks
0.1	Receptionists and information cl...
0.1	Computer, automated teller, and...
0.1	Maintenance and repair workers...
0.1	Office clerks, general
0.0	Stationary engineers and boiler...
0.0	Tellers
0.0	Ushers, lobby attendants, and ti...

Message. Employed leavers scatter widely; the largest block stays in education outside the classroom.

Policy. Skills remain in the education orbit, so re-entry is a realistic margin.

Leaving teaching, by age and route



Message. Leaving is U-shaped in age, and non-employment dominates both ends of the career.

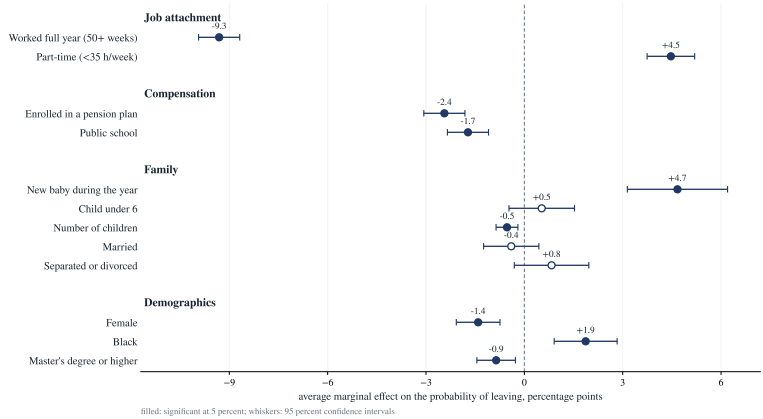
Policy. Early-career and pre-retirement exits call for different instruments.

The econometric specification

$$\Pr\big(\text{Leave}_{it}=1 \mid x_{it}\big) \;=\; \Phi\Big(\underbrace{\text{attach}'_{it}\beta_1}_{\substack{\text{full year,}\\ \text{part-time}}} + \underbrace{\text{comp}'_{it}\beta_2}_{\substack{\text{pension,}\\ \text{public school}}} + \underbrace{\text{fam}'_{it}\beta_3}_{\substack{\text{new baby, children,}\\ \text{marital status}}} + \underbrace{\text{dem}'_{it}\beta_4}_{\substack{\text{age, age}^2\text{, sex,}\\ \text{race, education}}} + \delta_t\Big)$$

- Probit on college-graduate teachers, pooled 2015–2024, with calendar-year effects δ_t; $n = 28{,}900$.
- Average marginal effects reported, in percentage points.
- Robust to logit and weighted LPM, state and year fixed effects, the full 1997–2024 window ($n = 85{,}430$), and the occupation-pair definition of leaving.

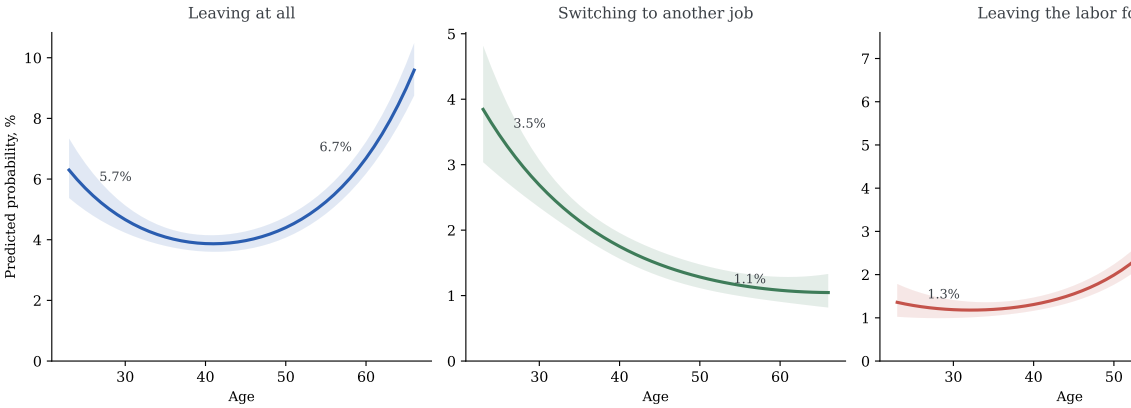
Average marginal effects on the probability of leaving



Message. Job attachment and compensation dominate; demographic differences are small.

Policy. The actionable levers are contractual: full-year contracts, pension coverage.

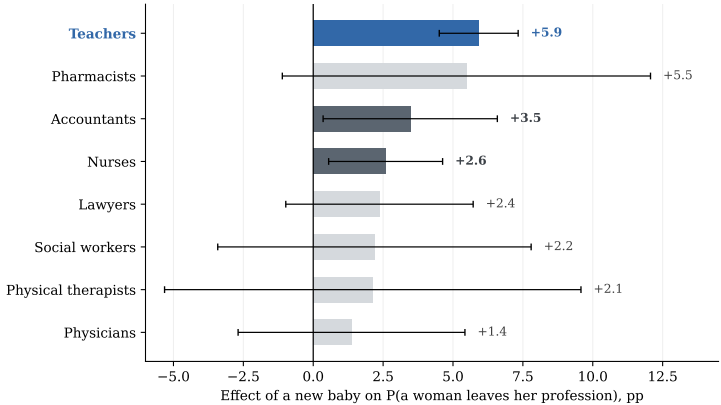
The age profile of leaving, by route



Message. Young teachers switch to other jobs; older teachers leave the labor force.

Policy. The two peaks of the U respond to different policies.

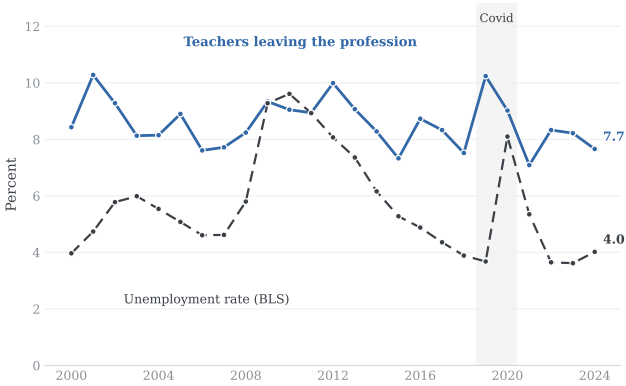
The effect of a new baby, by profession

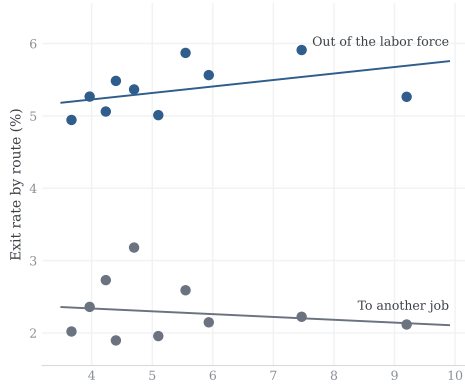


Message. A new baby raises a woman teacher's exit probability by 5.9 points, the largest effect among comparable professions.

Policy. Parental leave and childcare support target the largest single observable trigger.

Leaving teaching and the unemployment rate

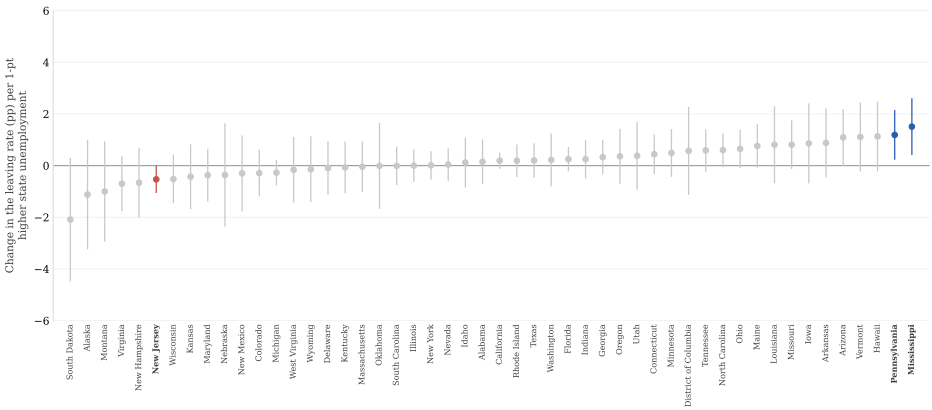




Message. Exits into non-employment rise with unemployment; job-to-job exits do not.

Policy. Downturns intensify the non-employment route; support should be countercyclical.

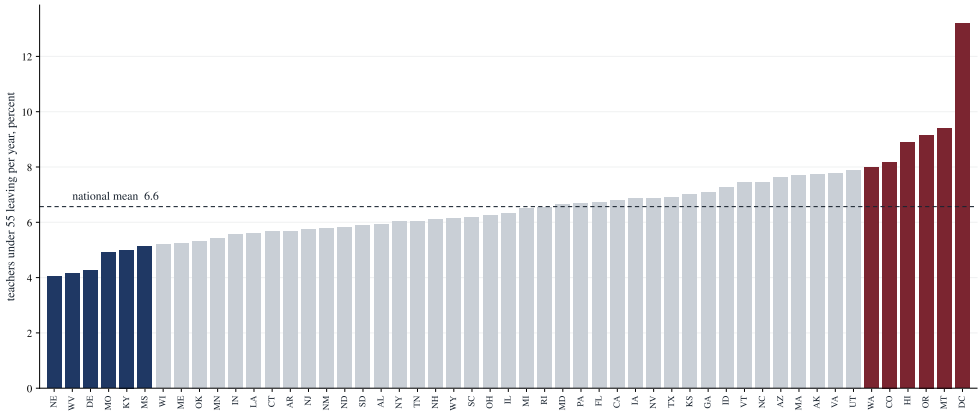
State-level cyclicality of leaving



Message. States differ in how strongly teacher exits respond to the local cycle.

Policy. National averages hide state-level exposure to downturns.

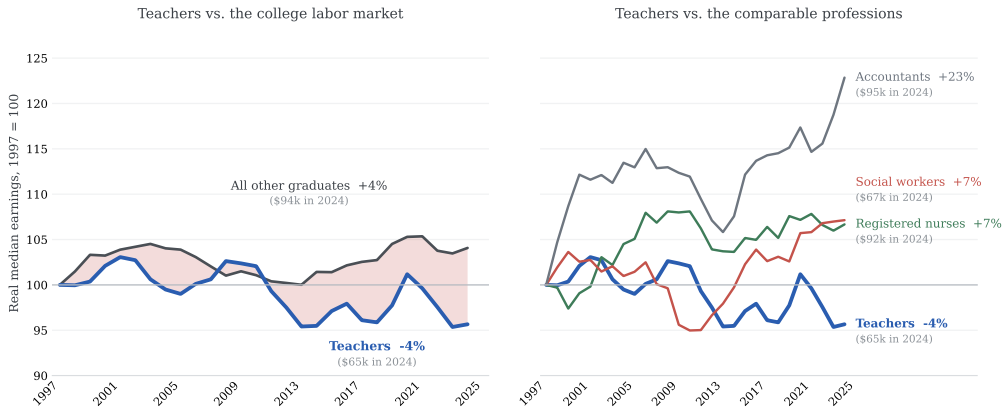
Attrition levels across states



Message. Mean attrition of teachers under 55 ranges from about 4 to 13 percent across jurisdictions.

Policy. State pay scales, pensions and institutions are first-order.

Real earnings growth by profession since 1997



Message. Teachers' real median earnings fell 4 percent since 1997 while other graduates gained.

Policy. The relative-pay gap compounds slowly; it is a stock, not a shock.

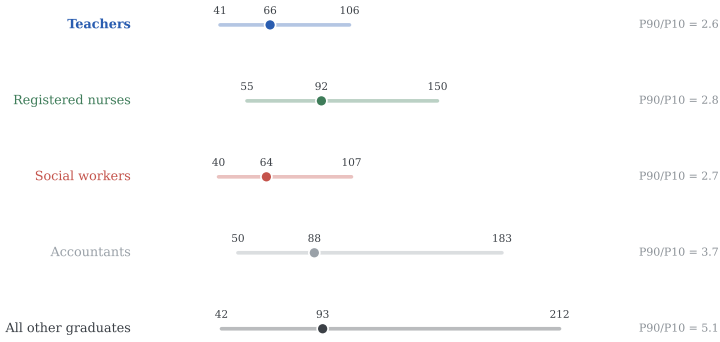
Leaving, by position in the teacher pay distribution



Message. Leaving concentrates sharply in the lowest earnings deciles, mostly part-year teachers.

Policy. Stabilizing the bottom of the distribution buys the most retention.

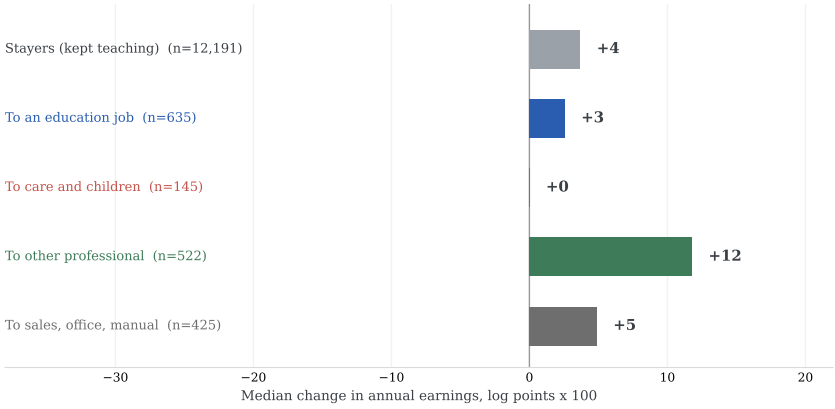
Pay compression: teachers and comparable professions



Message. Teacher pay is compressed: a 90/10 ratio of 2.6 against 5.1 for all other graduates.

Policy. Staying offers little earnings upside; career ladders would widen the right tail.

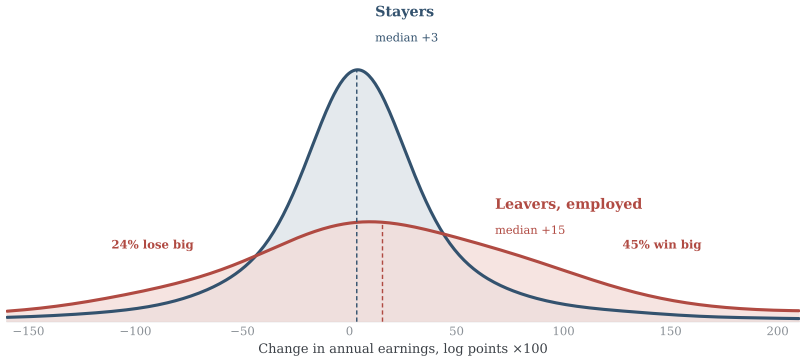
The earnings change of leaving, by destination



Message. Leavers who move to other professional jobs gain about 12 log points; moves near education gain little.

Policy. The outside option is real for those able to take it.

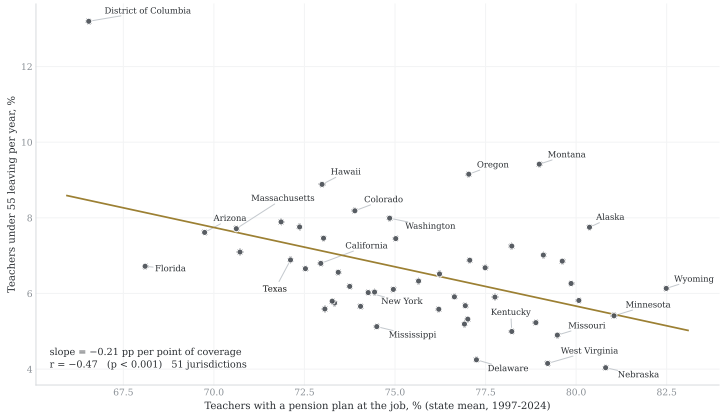
Earnings changes: stayers and leavers



Message. For employed leavers the earnings change is a gamble: a quarter lose big, nearly half win big.

Policy. Leaving buys risk, not a sure raise.

Pension coverage and leaving, across states



Message. Where pension coverage is higher, fewer teachers under 55 leave: 0.21 points per point of coverage.

Policy. Pension design is a measurable retention margin.

Conclusion

- A labour force survey can measure teacher attrition at the person level, yearly, over three decades, and it replicates the official picture of the workforce.
- The definition of a leaver is a first-order choice: the same records yield rates from 5 to 19 percent per year.
- What the data say: attachment, pay and pensions, and life-cycle events correlate with leaving; demographics much less.
- The design travels: the same construction is feasible in the labour force surveys of Italy, Germany, the UK, France, Spain, Brazil and Mexico.